Please give me a super meat boy like character controller in godot 4.4

Absolutely! I can help you set up a **Super Meat Boy–style character controller** in **Godot 4.4**. We'll aim for tight, responsive platforming with **fast movement, wall jumps, and precise air control**. I'll provide a **GDScript** example using `CharacterBody2D` (the recommended node for Godot 4 2D physics).

Here's a solid starting point:

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1. Node Setup

Your scene structure should look like this:

```
Player (CharacterBody2D)
├── Sprite2D
└── CollisionShape2D
```

Code

- `CollisionShape2D` : Set to a rectangle roughly matching your character.
- `RayCast2D` : Points downward for checking if you're grounded (optional, you can also use `is_on_floor()`).

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2. GDScript: PlayerController.gd

```
extends CharacterBody2D

# --- Movement variables ---
@export var move_speed: float = 300
```

```
@export var jump_force: float = 600
@export var gravity: float = 1800
@export var max_fall_speed: float = 1500
@export var air_control: float = 0.8

# --- Wall jump ---
@export var wall_jump_force: Vector2 = Vector2(400, -600)
@export var wall_slide_speed: float = 200

# Internal state
var velocity: Vector2 = Vector2.ZERO
var facing_right: bool = true

func _physics_process(delta):
    var input_dir = Input.get_action_strength("ui_right") - Input.get_action_stre

    # --- Horizontal movement ---
    if is_on_floor():
        velocity.x = input_dir * move_speed
    else:
        # Air control
        velocity.x = lerp(velocity.x, input_dir * move_speed, air_control)

    # --- Gravity ---
    if not is_on_floor():
        velocity.y += gravity * delta
        velocity.y = min(velocity.y, max_fall_speed)
    else:
        velocity.y = 0

    # --- Jumping ---
    if Input.is_action_just_pressed("ui_jump"):
        if is_on_floor():
            velocity.y = -jump_force
        elif is_on_wall():
            # Wall jump
            var wall_dir = get_wall_dir()
            velocity.x = -wall_dir * wall_jump_force.x
            velocity.y = wall_jump_force.y
```

```
    # --- Wall sliding ---
    if is_on_wall() and not is_on_floor() and velocity.y > wall_slide_speed:
        velocity.y = wall_slide_speed

    # --- Move the character ---
    velocity = move_and_slide(velocity, Vector2.UP)

    # --- Flip sprite ---
    if input_dir != 0:
        facing_right = input_dir > 0
        $Sprite2D.flip_h = not facing_right

# --- Helper function ---
func is_on_wall() -> bool:
    return is_on_wall_left() or is_on_wall_right()

func is_on_wall_left() -> bool:
    return test_move(global_transform, Vector2(-1, 0) * 1)

func is_on_wall_right() -> bool:
    return test_move(global_transform, Vector2(1, 0) * 1)

func get_wall_dir() -> int:
    if is_on_wall_left():
        return -1
    elif is_on_wall_right():
```

gdscript

```
    return 0
```

---

3. Notes on Behavior

- **Tight, snappy movement**: The `lerp` in the air gives a bit of control but keeps it tight.
- **Wall jumps**: Push away from walls naturally.
- **Gravity & fall speed**: Can be tuned to feel like Super Meat Boy (fast fall, strong jumps).
- **Sprite flipping**: Automatically flips the sprite depending on movement direction.

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4. Tuning Tips

- `move_speed` : Try **300–400** for SMB-like speed.
- `jump_force` : Around **600–700**.
- `gravity` : SMB is floaty but fast, try **1800–2200**.
- `wall_jump_force` : Tweak `x` and `y` to control wall jump distance and height.

If you want, I can also provide a **fully polished version with coyote time, jump buffering, and better wall detection**, which makes the controller feel exactly like Super Meat Boy.

Do you want me to do that next?

Fantastic please modify for variable jump height now

Love it — variable jump height is **essential** for that tight, Super Meat Boy–style feel.

We'll implement:

- ✅ **Short hop** (tap jump)
- ✅ **Full jump** (hold jump)
- ✅ **Faster fall when jump is released early**
- ✅ Works with wall jumps too

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- Increasing gravity when falling
- Applying extra gravity if the jump button is released while rising

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🔥 Updated Controller (Variable Jump Height)

Replace your script with this improved version:

gdscript

```
extends CharacterBody2D

# --- Movement variables ---
@export var move_speed: float = 350
@export var jump_force: float = 650
@export var gravity: float = 2000
@export var max_fall_speed: float = 1600
@export var air_control: float = 0.15

# --- Variable jump ---
@export var jump_cut_multiplier: float = 0.5   # How much upward velocity is kept
@export var fall_gravity_multiplier: float = 1.5  # Faster falling for snappier f

# --- Wall jump ---
@export var wall_jump_force: Vector2 = Vector2(450, -650)
@export var wall_slide_speed: float = 250

var facing_right: bool = true

func _physics_process(delta):
    var input_dir = Input.get_action_strength("ui_right") - Input.get_action_stre

    # --- Horizontal movement ---
    if is_on_floor():
        velocity.x = input_dir * move_speed
    else:
        velocity.x = lerp(velocity.x, input_dir * move_speed, air_control)

    # --- Gravity ---
    if not is_on_floor():
        # Apply stronger gravity when falling
        if velocity.y > 0:
            velocity.y += gravity * fall_gravity_multiplier * delta
        else:
            velocity.y += gravity * delta

        velocity.y = min(velocity.y, max_fall_speed)

    # --- Jumping ---
    if Input.is_action_just_pressed("ui_jump"):
```

```
        if is_on_floor():
            velocity.y = -jump_force
        elif is_on_wall():
            var wall_dir = get_wall_dir()
            velocity.x = -wall_dir * wall_jump_force.x
            velocity.y = wall_jump_force.y

    # --- Variable jump height (jump cut) ---
    if Input.is_action_just_released("ui_jump") and velocity.y < 0:
        velocity.y *= jump_cut_multiplier

    # --- Wall sliding ---
    if is_on_wall() and not is_on_floor() and velocity.y > wall_slide_speed:
        velocity.y = wall_slide_speed

    velocity = move_and_slide()

    # --- Flip sprite ---
    if input_dir != 0:
        facing_right = input_dir > 0
        $Sprite2D.flip_h = not facing_right


# --- Wall detection helpers ---
func is_on_wall() -> bool:
    return is_on_wall_left() or is_on_wall_right()

func is_on_wall_left() -> bool:
    return test_move(global_transform, Vector2.LEFT)

func is_on_wall_right() -> bool:
    return test_move(global_transform, Vector2.RIGHT)

func get_wall_dir() -> int:
    if is_on_wall_left():
        return -1
    elif is_on_wall_right():
        return 1
    return 0
```

🎮 What This Adds

Short Tap

If you quickly tap jump → small hop.

Hold Jump

If you hold jump → full height.

Faster Falling

The `fall_gravity_multiplier` makes the character fall faster than they rise — this gives that **Super Meat Boy snap**.

🎯 Recommended Feel Settings (Very SMB-like)

Try this combo:

```gdscript
move_speed = 380
jump_force = 700
gravity = 2200
jump_cut_multiplier = 0.4
fall_gravity_multiplier = 1.8
```

This gives:

- Very punchy jumps
- Aggressive falling
- Clean short hops
- Tight control

If you'd like, next I can add:

- ⭐ Coyote time
- ⭐ Jump buffering
- ⭐ Proper wall detection using RayCasts

- ⭐ Acceleration instead of instant movement
- ⭐ Dash system
- ⭐ Fully competitive-grade controller

Just tell me how hardcore you want this to feel 😈